

The University of the West Indies
Cave Hill Campus
Department of Economics
ECON 2025: Statistical Methods I
(Semester 1, 2026–2027)

Tutorial No.1
September 1, 2026

1. Consider the following set,

$$S = \{A, B, C, D\}$$

Two objects are to be selected from the set S .

- (a) List all the possible combinations of two objects that can be selected from the set S .
 - (b) Verify using the combination formula that the number of combinations of two objects that can be selected from the set S is equal to the number of combinations listed in part (a).
 - (c) List all the possible permutations of two objects that can be selected from the set S .
 - (d) Verify using the permutation formula that the number of permutations of two objects that can be selected from the set S is equal to the number of permutations listed in part (c).
 - (e) Clearly explain the difference between a combination and a permutation.
2. There are four routes A, B, C and D between a person's home and office. Route B can only be used travelling from home to office, while route C can only be used travelling from office to home.
- (a) Determine the routes available when travelling to work and from work to home using a tree diagram as well as the multiplication rule.
 - (b) Determine the number of possible round trips that can be made from home to office and back to home.
 - (c) How many round trips are possible if the person cannot use the same route in both directions?
 - (d) Suppose a fifth route E is added to the existing four routes. Determine the number of possible round trips that can be made from home to office and back to home.
3. Eight athletes compete in a 100-metre final. The first three athletes to cross the finish line will receive gold, silver and bronze medals respectively.
- (a) In how many ways can the medals be awarded to the athletes?
 - (b) In how many ways can all eight athletes finish the race from first through eighth place?
 - (c) If one particular athlete is guaranteed to finish in the top three, how many ways can the medals be awarded to the athletes?

- (d) Six persons are seated around a circular table. How many distinct circular arrangements are possible.
4. A ship contains 10 television sets, of which 3 are defective and 7 are non-defective. An inspector selects 4 sets.
- In how many ways can four sets be selected?
 - In how many selection are exactly 2 defective sets selected?
 - In how many selections are exactly 3 defective sets selected?
 - In how many selections are at least 2 defective sets selected?
 - Determine the number containing at most 1 defective set.
 - Verify part (e) by using the complement rule using the result from part (d).
5. A researcher has

$$N = 10$$

patients who are randomly grouped into two groups:

$$\text{Treatment} = 4, \quad \text{Control} = 6$$

- Determine the number of ways to select 4 patients for the treatment group.
- Show that,

$$\binom{10}{4,6} = \binom{10}{4}$$

- Use symmetry to show that,

$$\binom{10}{4} = \binom{10}{6}.$$

and explain why.

- Now suppose 10 students are placed in 3 tutorial groups of size 4,3,3. Determine the number of ways to assign the students to the three groups using the multinomial coefficient formula.
6. Consider the following binomial expression:

$$(x + y)^8$$

- Write the Binomial theorem for the above expression using summation notation.
- Use the Binomial theorem to expand the above expression.
- Evaluate each binomial coefficient and hence write the expansion in standard form.

(d) From your expansion, determine the coefficient of x^5y^3 .

(e) Identify the coefficients of x^8 , x^7y , x^6y^2 and x^4y^4 .

7. Construct Pascal's Triangle from

$$n = 0$$

through

$$n = 6$$

Using the triangle:

(a) Write the coefficients corresponding to

$$(x + y)^0, \dots, (x + y)^6$$

(b) Expand

$$(x + y)^6.$$

(c) Evaluate the following binomial coefficients:

$$\binom{6}{2}, \quad \binom{6}{3}, \quad \binom{6}{4}$$

(d) Verify the symmetry property of the binomial coefficients for

$$\binom{6}{2} = \binom{6}{4}.$$

8. The pro at a golf course stocks two identical sets of women's clubs, reordering at the end of each day (for delivery next morning) if and only if he has sold them both. Construct a tree diagram to show that if he starts on Monday with two sets of the club, there are altogether eight ways in which he can make sales on the first two days of the week.
9. A thermostat will call for heat 0, 1, or 2 times in a given night. Construct a tree diagram to show that there are altogether 10 different ways that it can turn on the furnace for a total of 6 times over 4 nights.
10. A student can study 0, 1, 2, or 3 hours for a test on any given day. Construct a tree diagram to show that there are altogether 13 ways in which the student can study at most 4 hours for the test on two consecutive days.

End of Tutorial 1